

Assignment: - Inheritance Javascript

A] Single Inheritance

Q1.Create two ES6 classes to demonstrate single inheritance

Problem Statement

1. **Parent class:** Teacher
 - Contains a method named `teach()` that prints a teaching-related message to the console.
2. **Child class:** MathTeacher
 - Inherits from Teacher using the `extends` keyword.
 - Adds its own method named `explainMath()` that prints a math-specific teaching message to the console.
3. Create an object of MathTeacher and call **both**:
 - The inherited method `teach()`
 - The child class method `explainMath()`

Expected Console Output

```
Teacher is teaching students  
Math Teacher is explaining Algebra
```

Q2. Create two classes to demonstrate Single Inheritance with Constructor:

1. **Parent Class: Vehicle**
 - Should contain a **constructor** that accepts brand and model of the vehicle.
 - Should have two methods:
 - `start()` → prints that the vehicle is starting.
 - `stop()` → prints that the vehicle is stopping.
2. **Child Class: Car**
 - Should inherit from Vehicle using the `extends` keyword.
 - Should have its own **constructor** that accepts brand, model, and fuelType.
 - Inside the constructor, use `super(brand, model)` to call the parent class constructor.
 - Should have a method `displayDetails()` that prints all the details: brand, model, and fuel type.
3. Create an object of Car and call:

- The parent methods (start(), stop())
- The child method (displayDetails())

Expected Console Output

```
Toyota Innova is starting...  
Car Details: Toyota Innova, Fuel: Diesel  
Toyota Innova is stopping...
```

B] Multilevel Inheritance

Q1. Create a program that demonstrates Multilevel Inheritance using the following hierarchy:

1. Base Class: Person

- **Constructor:** accepts name and age.
- **Method:** displayPerson() → prints "Name: <name>, Age: <age>".

2. Derived Class: Employee (inherits Person)

- **Constructor:** accepts name, age, and employeeId.
- Uses super(name, age) to call the parent constructor.
- **Method:** displayEmployee() → prints "Employee ID: <employeeId>".

3. Derived Class: Manager (inherits Employee)

- **Constructor:** accepts name, age, employeeId, and department.
- Uses super(name, age, employeeId) to call parent constructor.
- **Method:** displayManager() → prints "Department: <department>".

4. Task

- Create an object of Manager.
- Call all three methods:
 - displayPerson() (from Person)
 - displayEmployee() (from Employee)
 - displayManager() (from Manager)

Expected Console Output

```
Name: Akanksha, Age: 28
```

Employee ID: E102
Department: IT

C]Hierarchical Inheritance

Q1.Create a program to demonstrate Hierarchical Inheritance using the following hierarchy:

1. Base Class: Shape

- **Method:** area() → prints "This is the area method for shapes".

2. Derived Class: Circle (inherits Shape)

- **Constructor:** accepts radius.
- **Method:** area() → calculates and prints "Area of Circle: <calculated value>".

3. Derived Class: Rectangle (inherits Shape)

- **Constructor:** accepts length and width.
- **Method:** area() → calculates and prints "Area of Rectangle: <calculated value>".

4. Derived Class: Triangle (inherits Shape)

- **Constructor:** accepts base and height.
- **Method:** area() → calculates and prints "Area of Triangle: <calculated value>".

5. Task

- Create objects of Circle, Rectangle, and Triangle.
- Call the area() method for each object.

Expected Console Output

```
Area of Circle: 78.5  
Area of Rectangle: 60  
Area of Triangle: 16
```